SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 9: Travelling Salesman Problem

CODE:

```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
best_cost = float("inf")
best_path = None

for route in permutations(range(1, n)):
    path = (0,) + route + (0,)
    cost = sum(
        graph[path[i]][path[i+1]]
        for i in range(n)
    )
```

```
    if cost < best_cost:
        best_cost = cost
        best_path = path

print("Best route:", best_path)
print("Minimum cost:", best_cost)
```

OUTPUT:

